

INNOVATIVE MINING SYSTEMS

Intellectual Property Rights and Development

By

**Prof. Santosh Kumar Behera, Assistant Professor,
Mining Engineering**



**Indian Institute of Technology (Indian
School of Mines), Dhanbad**

What is Intellectual Property (IP) ?

- Intellectual Property (IP) is a **bundle of rights** that protects **applications of ideas and information** that have **commercial value**.
- IP rights give creators certain **exclusive rights** over the knowledge and information they create, such as an invention or a work or a mark or a design, to **prevent others using it** without permission.

What is Development ?

- “Development” means an “event constituting a new stage in a changing situation” (OED)
- Means improvement
- A multi-dimensional concept in its nature
- Includes , *inter alia*, economic and social development
- Economic development – increasing productivity and providing more resources
- Social Development
 - Cultural Development; Better Entertainment

Development and IP

- Both economic and social development need improved/better technologies and new/original creations in the cultural field
- Development of Technologies need investment
- Copying new innovations and creations is a disincentive for investment
- IPRs ensure certain rights for innovators and creators; Others cannot copy without permission
- Exclusive rights contribute to economic gains
 - Case of Toyota
- Attracting investment

Rationale of Intellectual Property

- To incentivise innovation and creative outputs
- A transparent mechanism for revelation
 - Contract between the State and Individual
- A transparent and democratic system for rewarding innovation and creativity

What does IPR do?

- Knowledge becomes effective by **entering the market**
- In market conditions only knowledge or expressions becomes **source of income**
- Market demands **control over the inventions** and creative outputs
- **IPRs provide this control**
- IPRs are of great significance in the current **knowledge driven global economy.**

Major Intellectual Property Rights

- **Copyright** and Related Rights
- Industrial Property
 - **Patents**
 - **Industrial Designs**
 - Trademarks
 - Geographical Indications
 - **Layout Designs/Topographies Integrated Circuits**
 - Trade Secrets
 - Protection of New Plant Varieties

Major IPR Conventions

- Paris Convention for the Protection of Industrial Property 1883
- Berne Convention for the Protection of Literary and Artistic Works 1886
- Madrid Agreement Concerning the International Registration of Marks 1891
- International Convention for the Protection of New Varieties of Plants 1961
- Patent Cooperation Treaty 1970
- Madrid Protocol 1989
- Agreement on Trade Related Aspects of Intellectual Property Rights 1994
- WIPO Copyright Treaty 1996
- WIPO Performances and Phonograms Treaty 1996

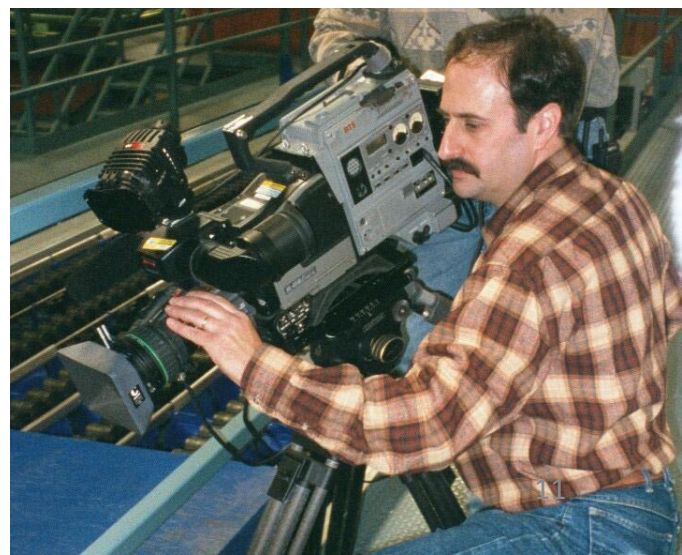
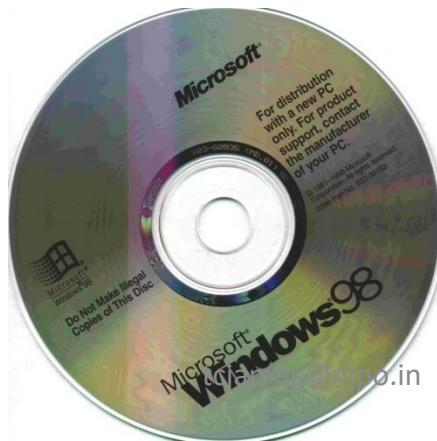
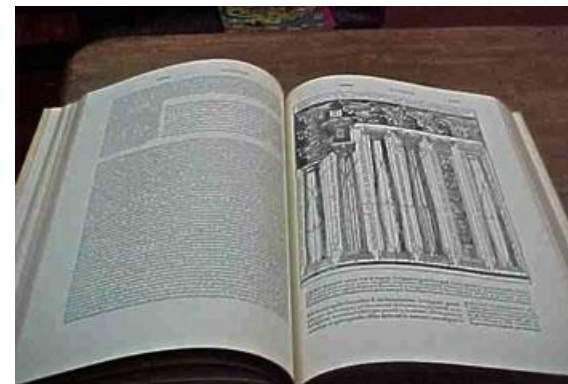
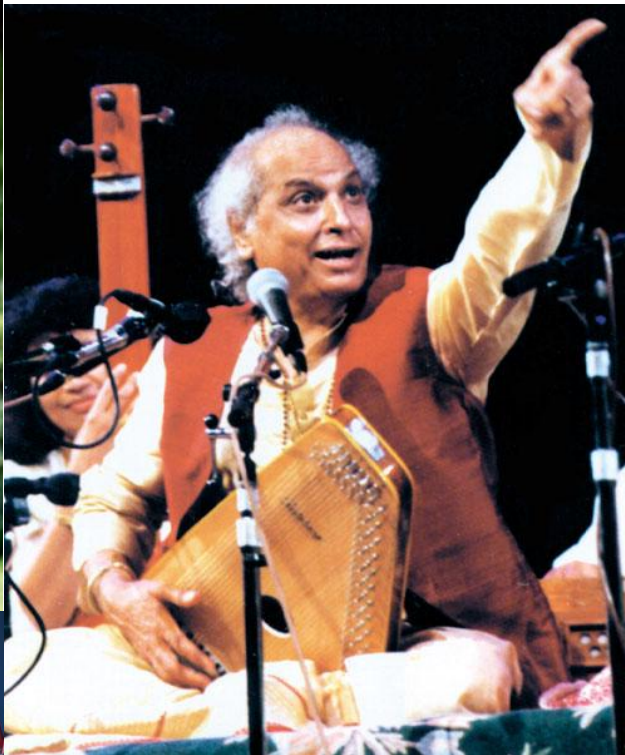
Major International Organisations in IPRs

- World Intellectual Property Organisation
- World Trade Organisation
- United Nations Educational, Scientific and Cultural Organisation

IP Laws of India

Act	Department
Copyright Act, 1957 (last amended in 2012)	Higher Education
Patents Act, 1970 (last amended in 2005)	Industrial Policy & Promotion
Designs Act, 2000	Industrial Policy & Promotion
Trademarks Act, 1999	Industrial Policy & Promotion
Geographical Indications of Goods (Registration and Protection) Act, 1999	Industrial Policy & Promotion
Semi-conductor Integrated Circuits Layout Design Act, 2000	Information Technology
Protection of Plant Varieties and Farmers' Rights Act, 2001	Agriculture and Cooperation

Copyright



What is Copyright?

- Copyright is a legal term describing rights given to creators for their literary and artistic works

Scope of Copyright

- Original Literary, Dramatic, Musical and Artistic Works
 - Work: Ideas expressed in material form
 - Literary or artistic quality not a criterion, but artistic craftsmanship needed for artistic works
 - No copyright in ideas or facts but only in expression
- Cinematograph Films
- Sound Recordings

**COPYRIGHT INDUSTRY A MAJOR CONTRIBUTOR OF
GDP OF A COUNTRY**

Scientists and Copyright

- Both Creators and Users
- Research Papers, Articles, Books, Artistic Designs, Charts, Graphs
- Computer software, Databases
- Lectures, Speeches
- Educational films

Rights

- Two Kinds of Rights

- Moral Rights

- ❖ To protect personality of author

- Economic Rights

- ❖ To bring economic benefits

Moral Rights

- Right of **Authorship**
- Right of **Integrity**
 - Against Distortions
 - Digital Manipulation
 - Morphing
 - No Right for Display
- Inalienable Rights

Economic Rights

- Right of Reproduction
 - Making copies e.g. an edition of a novel
 - Storage in computer memory
- Right of Distribution/Issuing Copies
 - Digital Distribution
- Right of Communication to the Public
 - Public Performance
 - Internet Communication

Economic Rights

- Adaptation Rights
 - Conversion into another form e.g. literary to drama
 - Abridgement
 - Picturizations, comic formats
- Right to make a cinematograph film or sound recording
- Translation Rights

Ownership of Rights

- Literary – Author
- Drama – Dramatist
- Music – Composer
- Artistic work – Artist e.g. Painter, sculptor, architect
- Photograph – Photographer
- Author of Computer Programme – Person who causes the work to be created
- Cinematograph film – Producer
- Sound Recording - Producer

Author as Owner of Rights: Exceptions

- In the course of employment – employer
- Employment by newspaper, magazine – employer has publishing right; other rights with author
- Photograph, painting, cinema made for valuable consideration – person who pays money

Author as Owner of Rights: Exceptions

- Lecture delivered in public – Person delivering
- Government Work – Government
- Public Undertaking Work – public undertaking
- Work of International Organization – International Organization

Securing Copyright

- Formality free protection
- Voluntary Registration
- Registration does not as a matter of law establish that what is registered is in fact and in law copyrightable subject matter
- Only one office that is in New Delhi

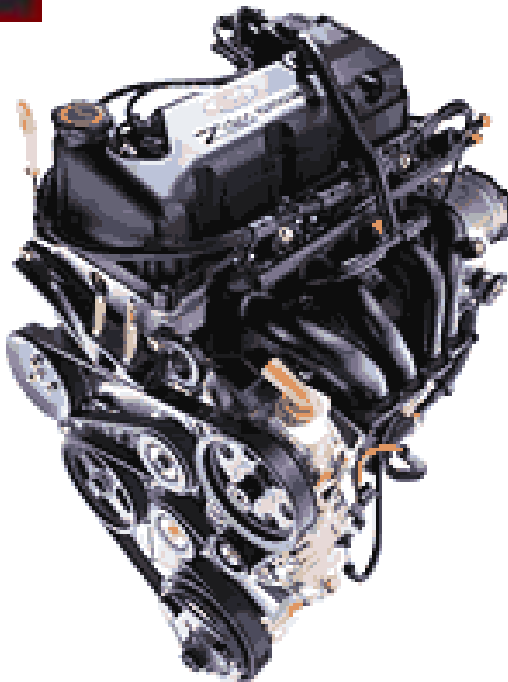
What is Not Infringement?

- Fair dealing for private use, research, criticism, review, reporting
- Judicial proceedings
- Work of legislature
- During course of instructions and examination
- Reproduction of public material
- Use in religious or official ceremonies including marriage processions

Duration of Copyright

- Literary, dramatic, Musical and Artistic Works published during life time of author: **Life + 60 years**
- All Other Works: **60 years** from date of publication
 - Posthumous, Anonymous Works
 - Works of Government and Organizations
 - Cinema and Sound Recording

Patents



Patents and Scientists

- Both Creator and User
- Secure one's own invention
 - *“Publish and Perish Or Patent, Publish and Prosper”*
 - Needs to obtain licence for using a patented invention or process
- Patent Database contains latest advancements in each field of technology.

What is a Patent?

- A patent is an exclusive right **granted** for an invention, which is a product or a process that provides a new way of doing something, or offers a new technical solution to a problem
- The **limited monopoly right** granted by the state enables an inventor to prohibit another person from manufacturing, using or selling the patented product or from using the patented process, without permission.
- Period of Patents - **20 Years subject to annual maintenance**

What Can Be Patented?

Inventions in all fields of technology, whether **products** or **processes**, if they meet the **criteria** of

- Novelty;
- Non-obviousness (inventive step);
- Industrial application (utility).

Conditions of Patentability

- **Novelty**: Invention **not known** to public prior to claim by inventor
 - e.g. **wound healing properties of turmeric**
- **Inventive Step**: Invention would **not be obvious** to a person with ordinary skill in the art
 - e.g. **aluminium frame for wooden frame**
- **Industrial Application**: Invention can be made or used in any **useful**, practical activity as distinct from purely intellectual or aesthetic one.

Inventions – Not Patentable

An Invention Which is frivolous or which claims anything obviously contrary to the well established Natural Laws e.g.

- Machine giving more than 100% performance
- Perpetual motion machine

Inventions – Not Patentable

A machine whose primary or intended use **or** commercial exploitation of which could be contrary to Public order **or** morality **or** which causes serious prejudice to human, animal or plant life or health or to the environment:

Device for house-breaking

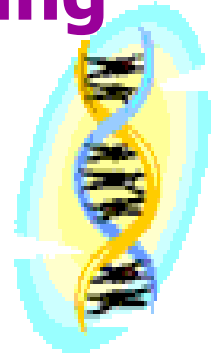
Biological warfare material or device

Terminator gene technology

Inventions – Not Patentable



Mere Discovery of a Scientific Principle
or Formulation of an Abstract Theory
or discovery of any living thing **or non–living**
substance occurring in nature



- Discovery** adds to the human knowledge by disclosing something ,not seen before, **whereas,***

- Invention** adds to human knowledge by suggesting an action resulting in a new product or new process*

- e.g.** Archimedes Principle, Superconducting Phenomenon as such – not patentable , **However,***

- An apparatus /method for technological application may be patentable*

Inventions – Not Patentable



The mere discovery of a new form of a known substance which does not result in the enhancement of the known efficacy of that substance OR

the mere discovery of any new property or new use for a known substance OR of the mere use of a known process, machine or apparatus, unless such known process results in a new product or employs at least one new reactant.



Inventions – Not Patentable

A substance obtained by mere admixture resulting only in the aggregation of the properties of the components thereof **Or** a process for producing such substance

For example, **Not patentable-**

- Paracetamol (Antipyretic) +Brufen (analgesic) = A drug (antipyretic & analgesic)
- A mixture of sugar and some colourants in water to produce a soft drink is mere admixture

But, a mixture resulting into synergistic properties of mixture of ingredients however, may be patentable

e.g Soap, Detergents, lubricants ,etc

Inventions – Not Patentable

Mere arrangement or re-arrangement or duplication of known devices, each functioning independently of one another in a known way

for example -.

- *A Bucket fitted with torch,*
- *An Umbrella with fan*
- *A Clock and radio in a single cabinet*
- *A flour-mill provided with sieving*



Inventions – Not Patentable

Method of Agriculture or Horticulture

- e. g. Cultivation of algae ,
Producing new form of a known plant,
preparation of an improved soil

However, Agricultural Equipments
are patentable



Inventions – Not Patentable

Any process for medicinal, surgical, curative, prophylactic, diagnostic, therapeutic or other treatment of human beings

or

a similar treatment of animals to render them free of disease **or** to increase their economic value or that of their products

For example:

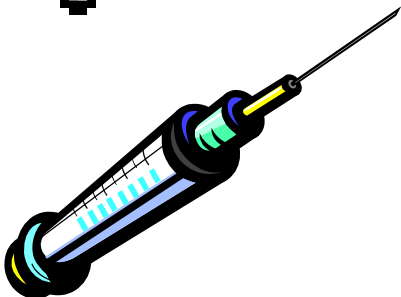
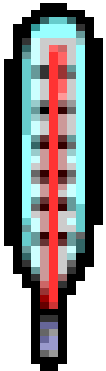
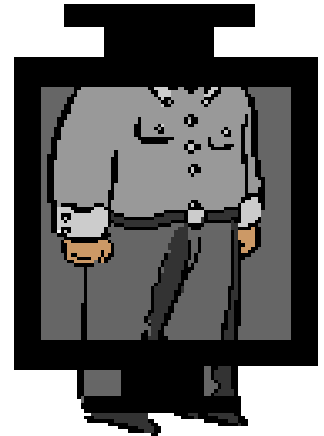
- Removal of cancer tumour
- Removal of dental plaque and cavities,
- Surgical processes,
- Processes relating to therapy,
- Method of vaccination,
- Blood transfusion



However,

- Method performed on tissues or fluids permanently removed from the body
- Surgical, therapeutic or diagnostic Apparatus or instrument

are not excluded from patentability



Inventions – Not Patentable

Plants & animals *in whole or any part thereof*, *other than* micro-organisms, *but* including seeds, varieties and species *and* essentially biological process for production or propagation of plants & animals

For example,

Clones and new varieties of plants:

- Not patentable

A process for production of plants or animals if it consists entirely of natural phenomena such as crossing or selection”- essentially biological
- not patentable



- **Inventions – Not Patentable**

- A literary, dramatic, musical or artistic work or any other aesthetic creation including cinematographic work and television productions

These subject-matters fall under the copy-right protection



A mere scheme or rule or method of performing mental act or method of playing game

Examples –

- Method for solving a crossword puzzle, Method of learning a language, Method of teaching /learning
- But Novel apparatus for playing game or carrying out a scheme – *patentable*



NOT PATENTABLE

Presentation of information

Examples –

Any manner or method of expressing information , whether by spoken words, Visual display, symbols ,diagrams or information recorded on a carrier

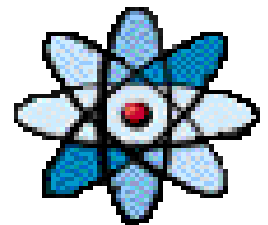
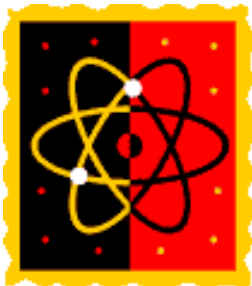
Topography of integrated circuits.

An invention which, in effect, is the

Traditional Knowledge or an aggregation or duplication of known properties of traditionally known component or components



- Traditional Knowledge is already in public domain, and hence, ***not patentable***
 - However, any value-addition using Traditional Knowledge leading to a new process or product, possessing novelty, inventive step and industrial applicability, can be ***patentable***
-
- **Inventions relating to atomic energy falling within the section 20(1) of the Atomic Energy Act, 1962.**



Rights of a Patentee and Exceptions

- **Exclusive right** to prevent third parties from the act of making, using, offering for sale, selling or importing the patented invention
- **EXCEPTIONS**
- **Government Use**
 - A **medicine or drug for distribution in government hospitals/dispensaries**
 - Importation of machines for own use
 - Use of process patent for own purpose
- For **research or experimental purposes** or for **teaching**
 - E.g. A teacher in an engineering college can depict, explain and show the functioning of patented invention to his pupils .

Different Ways Of Dealing With An Invention

- Make it public for free use by public (like publishing in the journal)

Or

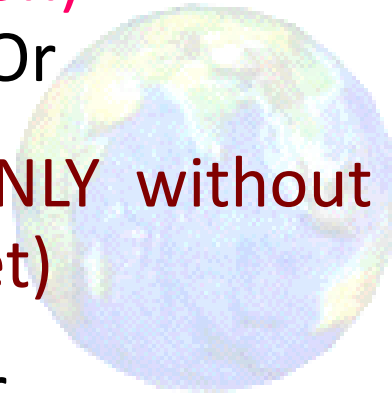
- Work the invention in **SECRECY** without **PATENTING** it (like coco-cola composition)

Or

- Work the invention **OPENLY** without **PATENTING** it (directly put it in the market)

Or

- **EXPLOIT** the invention on the basis of a **PATENT** (like Rank Xerox)



Patents Not Only For Major Technological Breakthroughs ...

- Stay-on-tabs (also called colon tabs) were invented by Daniel F. Cudzik of Reynolds Metals in Richmond, Virginia, in 1975.



Patents Not Only For Major Technological Breakthroughs ...

- Bottle cap with combination lock
Invented by Douglas Taylor Hamer
(US) patented in 2009



Business Built on Patents ...

- In 1896, Sakichi Toyota obtained patent for a power loom
- In 1909, invented an automatic loom and obtained several patents
- In 1924, Toyota Type G automatic loom introduced
- Kiichiro Toyota, son, gave exclusive rights to manufacture and sell in all countries except Japan, China and US to Platt Brothers for Pounds 100,000
- That was invested in R & D for an automobile company.

Why do You need “Patent Information”

- **SIZE OF THE RESOURCE –ENORMOUS AND WIDE** Every area of technology is covered.

- **80% NOT PUBLISHED ELSEWHERE**

- **FIRST PUBLICATION:**Inventions disclosed in patents well before being published in any other type of document

Invention	Patent published	First publication in any form
-----------	------------------	-------------------------------

Punch card	1889	1914
------------	------	------

Television	1923	1928
------------	------	------

Jet engine	1936	1946
------------	------	------

- **EXPIRED PATENTS : FREE USE ;**

- **TO AVOID REDUNDANT RESEACH** Use the technology given in patent specification as a stepping stone.

Patent Offices

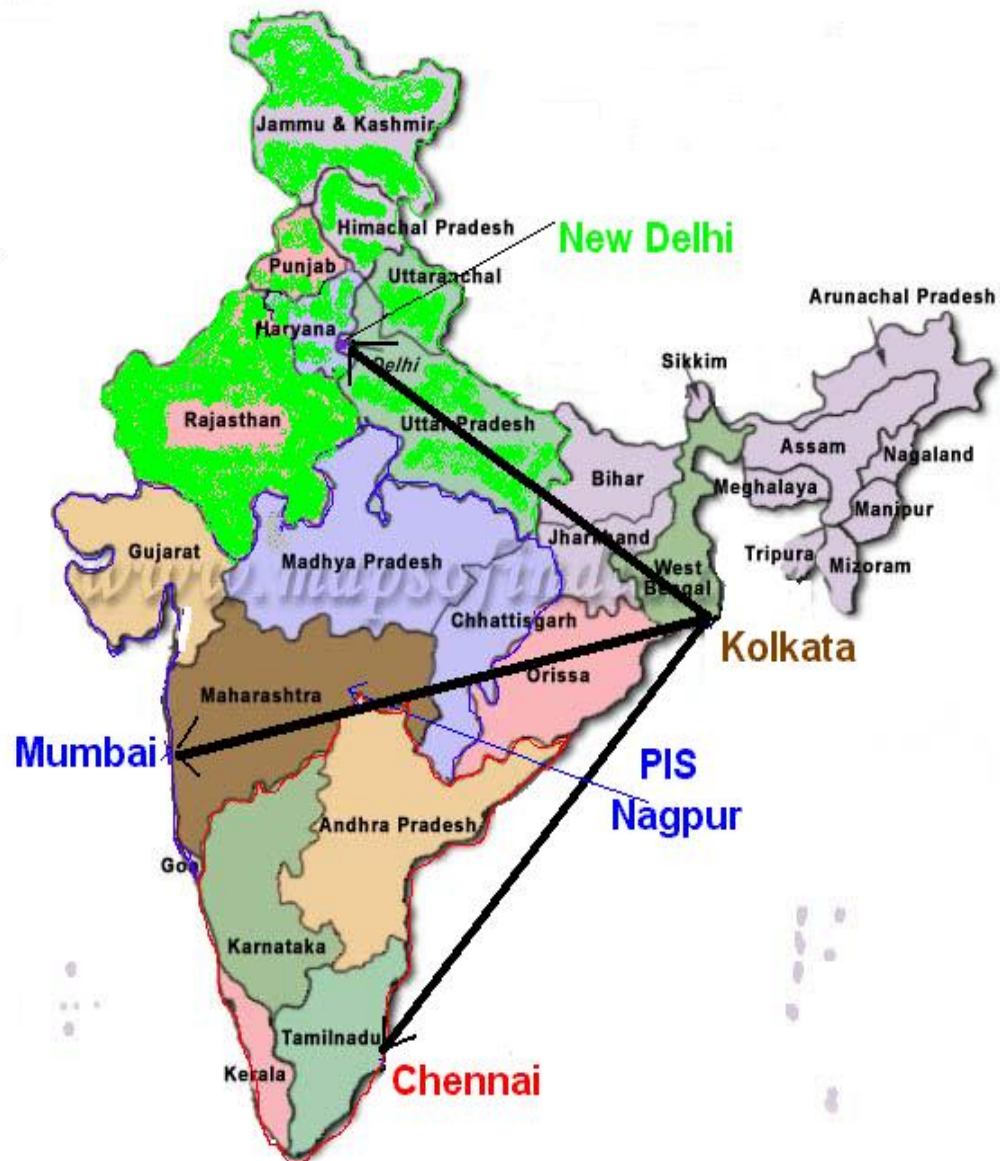
- Headquarters in **Kolkata**
- Regional offices in **Delhi, Mumbai** and **Chennai**
- Applications to be submitted to the office concerned
- **E-filing** facility available



www.ipindia.nic.in

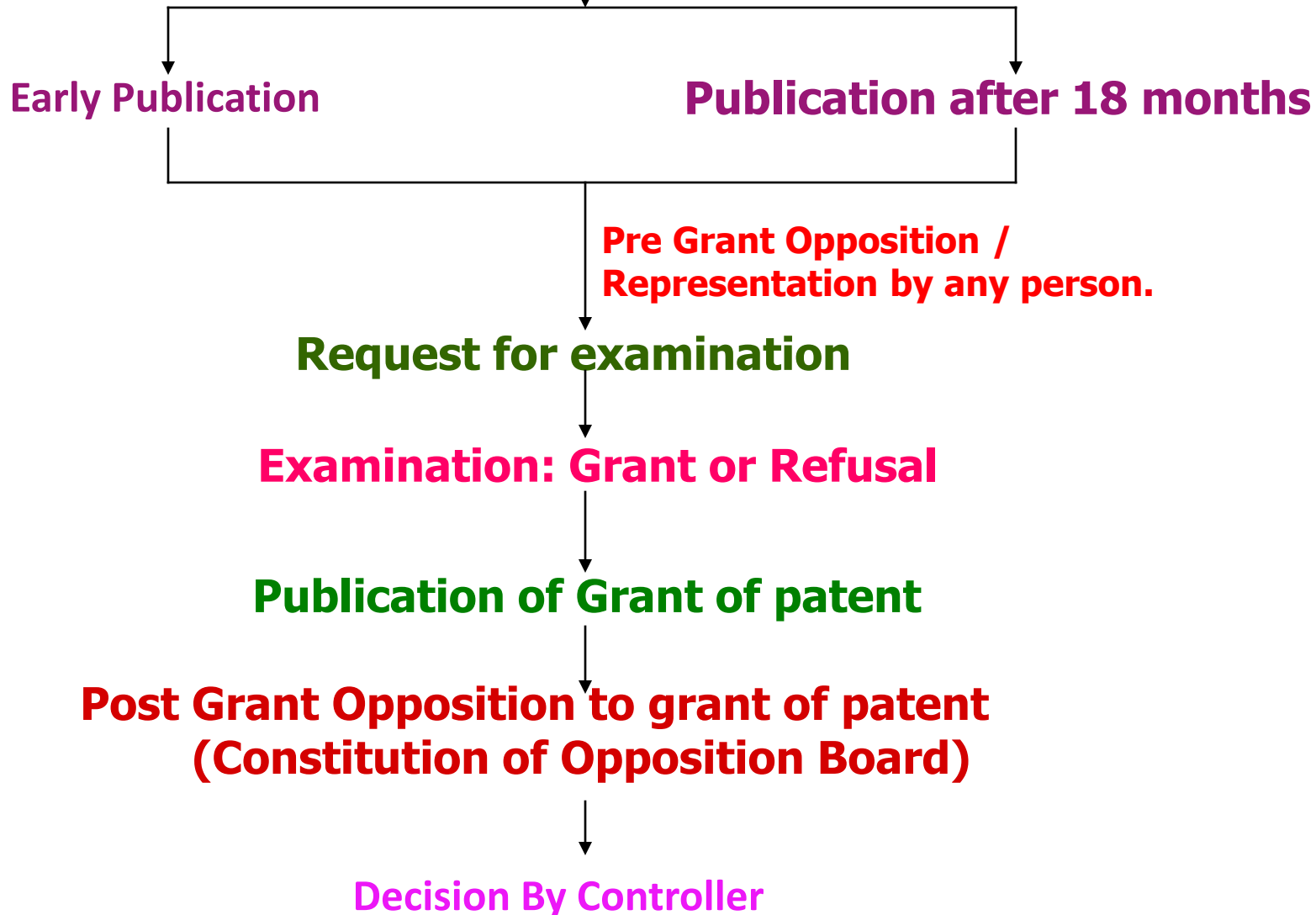


Jurisdiction of offices



Patent Grant Procedure

Filing of patent application



(10)



Europäisches Patentamt
European Patent Office
Office européen des brevets



(11)

EP 0 790 731 A1

(12)

EUROPEAN PATENT APPLICATION

(43) Date of publication:
20.08.1997 Bulletin 1997/34

(51) Int. Cl.: **H04M 1/72, H01M 2/10**

(21) Application number: **B610228B.3**

(22) Date of filing: **15.02.1996**

(81) Designated Contracting States:
DE FR GB IT SE

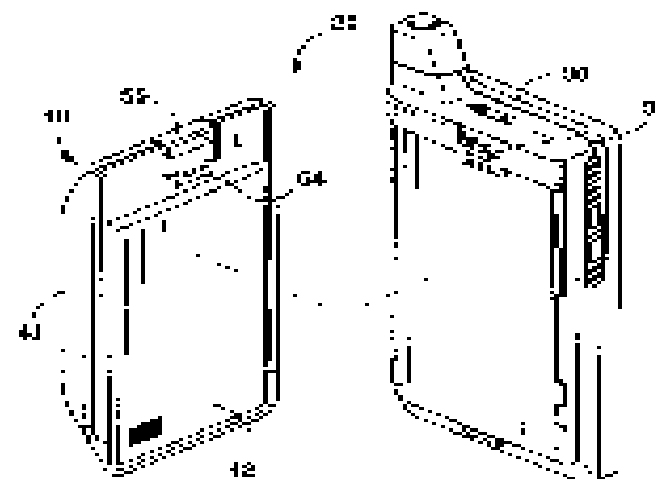
(72) Inventor: **Hsu, Cheng-Ming**
Taipei (TW)

(71) Applicant: **Hsu, Cheng-Ming**
Taipei (TW)

(74) Representative: **Wöring, Jentschura & Partner**
Postfach 22 14 43
80604 München (DE)

(51) Clasp structure for cellular phone and battery set

(57) A clasp structure of cellular phone and its battery set is consisted by one cellular phone and one battery set. The battery set is comprised by one case and battery base. On the battery case has one clasp component which uses elasticity of spring component to buckle and to separate the cellular phone and the battery set. When the battery exhausted the cover, only pushes the clasp component to make the battery set depart from the cellular phone; thus, not only easy to separate, but also the clasp component will not break. User can save lots of money and easy operate that can increase the convenience to users.



The
United
States
of
America



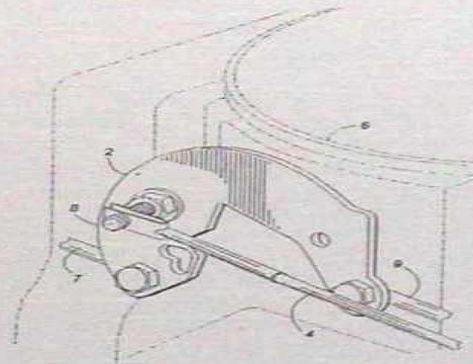
UNITED STATES PATENT

Granted on October 15, 2002

Randy Bouchillon
INVENTOR

6,463,826

AUTOMOBILE TRANSMISSION CABLE DEVICE



A carburetor to transmission cable linkage for Chrysler Torqueflight® transmissions found on carbureted, rear wheel drive Chrysler cars manufactured since 1962. A throttle lever adapter provided by the invention provides a connection of the cable to the carburetor. The cable is also mounted to the transmission. The opposite end of the cable is mounted to the throttle lever of the carburetor. The throttle lever adapter has a structure which allows it to be universally mounted to, and operable with, Carter, Edelbrock, and Holley carburetors, of various models, as used with Chrysler cars.

The Director of the United States Patent and Trademark Office has received an application for a patent for a new and useful invention. The requirements of law have been complied with, and it has been determined that a patent on the invention shall be granted under the law. Therefore, this

United States Patent

Grants to the person(s) having title to this patent the right to exclude others from making, using, offering for sale, or selling the invention throughout the United States of America or importing the invention into the United States of America for the term of this patent, subject to the payment of maintenance fees as provided by law.

James D. [Signature]
Director of the U.S. Patent and Trademark Office

SPECIFICATION
No. 1. 2nd January 1912.

(NOTE.—The application for a patent for an invention of which this is the specification was accepted on **5th February 1912**. The acceptance is advertised in the next issue of the "Gazette of India." Within three months from the date of the advertisement, notice of opposition to the grant of the patent may be given under § 9 of the Indian Patents and Designs Act, 1911, and Rule 14 of the Indian Patents and Designs Rules, 1912.)

**IMPROVED PROCESS AND APPARATUS FOR STERILISING AND
PURIFYING LIQUIDS AND REGENERATING PERISHABLE
LIQUIDS AND MATERIALS.**

**DOUGLAS LEICESTOR VERNON BROWNE AND JEAN MASSON, BOTH
OF 38, RUE DES PERCHAMPS, PARIS, FRANCE.**

The following specification particularly describes and ascertains the nature of the invention and the manner in which the same is to be performed.

The ordinary sterilising processes have for their essential object to kill the bacteria or other micro-organisms contained or generated in the liquids to be treated owing to the changes which may be produced in the medium by the development of those organisms (as in fermentations for example). The sterilising process must also, if it is desired to eliminate the suspended matter (such as dust or other particles, dead germs, etc.), be completed by filtering the treated liquid, for instance by means of a Chamberland porcelain apparatus. 5

Among sterilising processes there may be mentioned those processes employing chemical agents, that is to say, processes requiring for their application the use of reagents which modify by their mere presence the nature or properties of the medium to be sterilised, and process employing physical agents, such as sterilisation by heat, ozone, the ultra-violet rays or other invisible emanations of the spectrum. 10

In the last-mentioned processes, a source of heat, ozone, etc., is utilized, working externally and independently of the medium, and producing physical sterilising agents which are introduced into the liquid in the same manner as are the chemical agents and which are intended by acting through the liquid 15

Industrial Designs



What is an Industrial Design?

- An industrial design is the ornamental or aesthetic aspect of an article which must **appeal to the eye**. The design may consist of three-dimensional features, such as the shape or surface of an article, or of two-dimensional features, such as patterns, lines or colour

What is a Registrable Design?

- ‘Design’ means only the features of shape, configuration, pattern, ornament or composition of lines or colours *applied* to any article whether in two dimensional or three dimensional or in both forms, *by any industrial process* or means, whether manual, mechanical or chemical, separate or combined, which in the finished article *appeal to* and are judged solely by *the eye*.

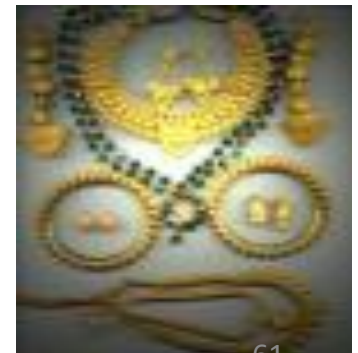
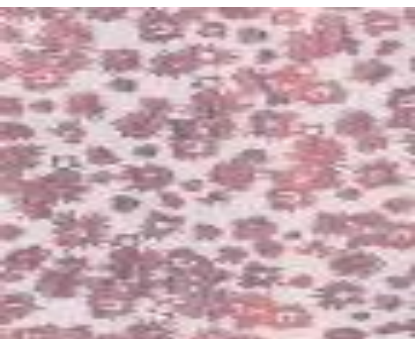
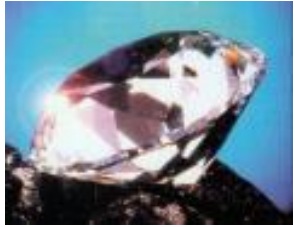
Consumer Products



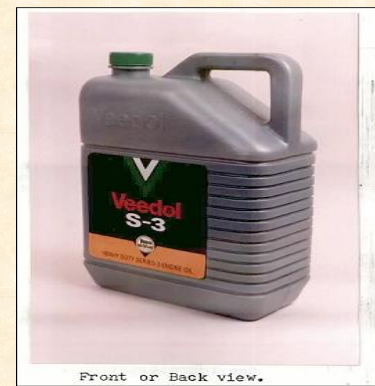
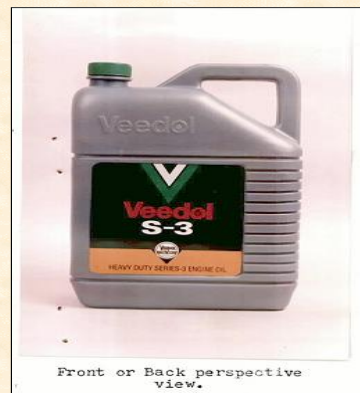
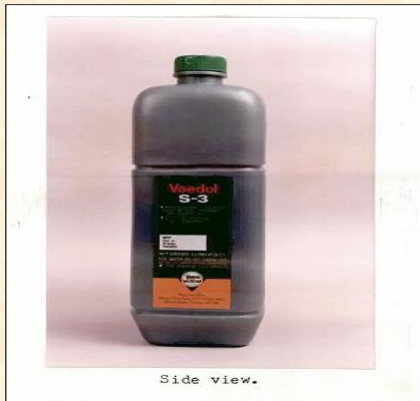
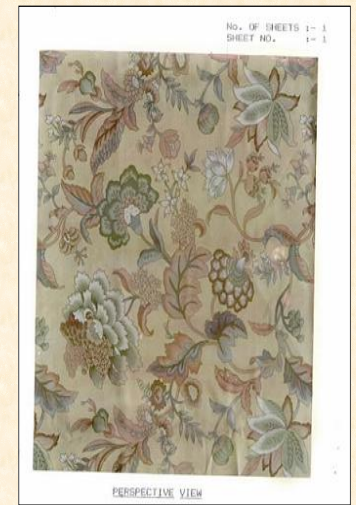
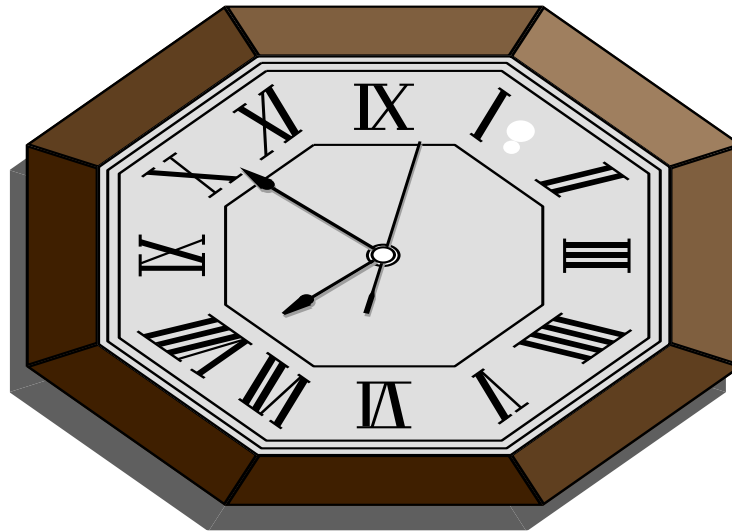
Pharmaceutical Products



Textile & Jewellery



DESIGNS



Design Examples



Rights of the Registered Proprietor

- Exclusive right to apply a design to any article in the class in which the design is registered
- Duration of protection is 10 years extendable by 5 years..

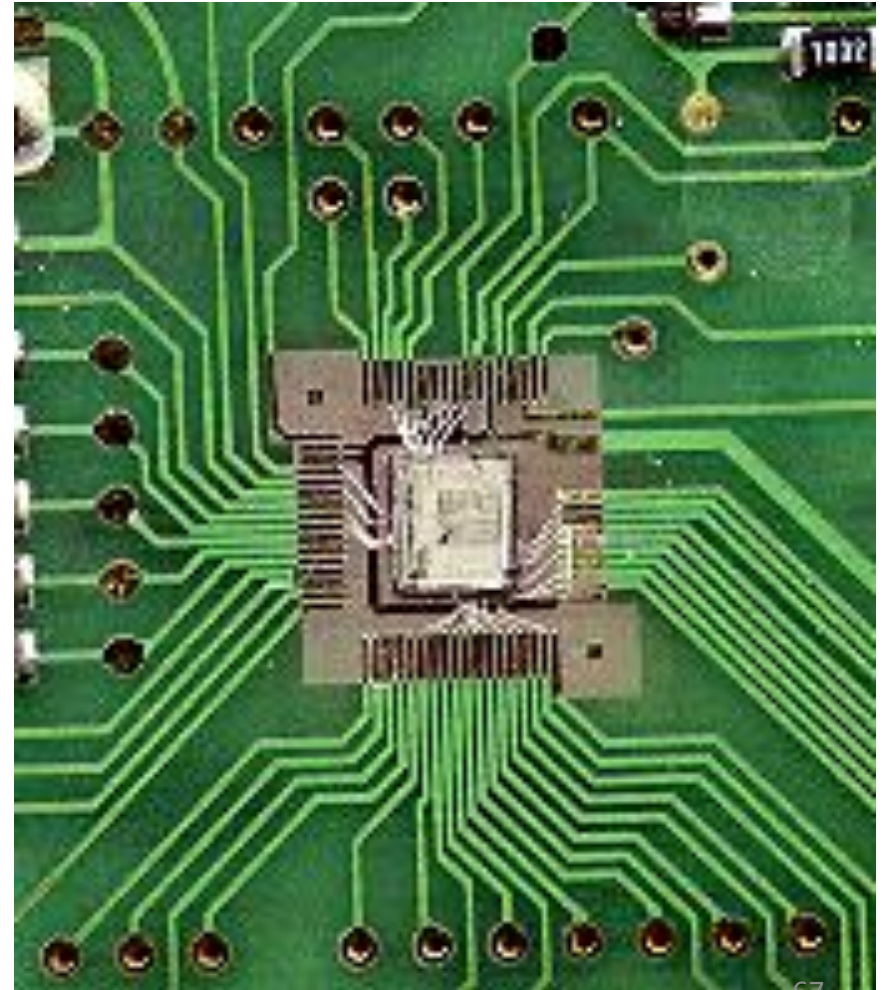
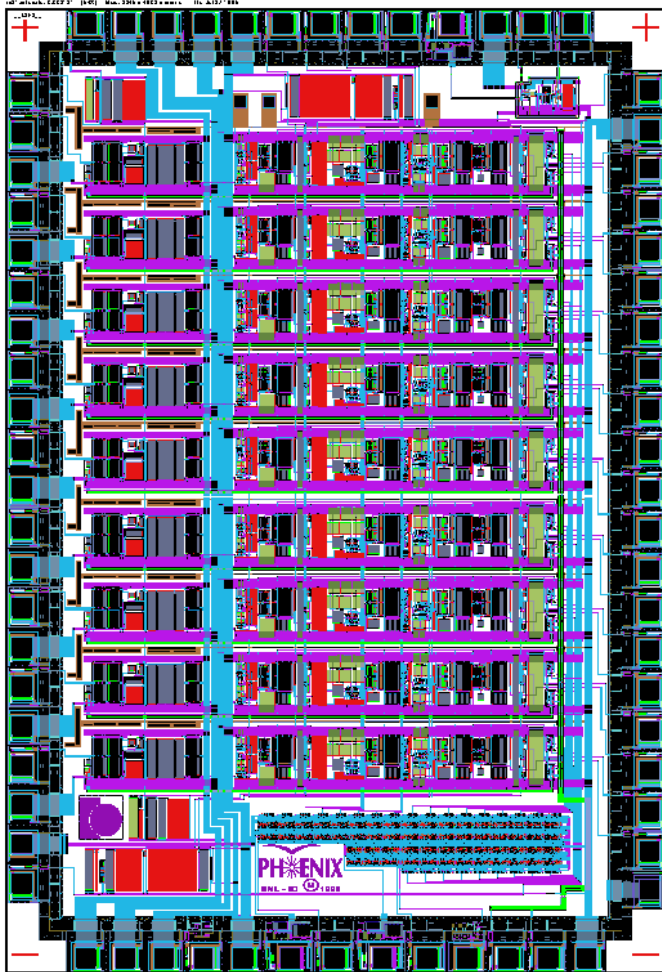
What is Excluded?

- Any **mode or principle of construction** or anything which is in substance *a mere mechanical device*
- Trade mark
- Property mark
- Artistic work

Registration

- Only One Office
- Design Wing of Patent Office in Kolkata
- Application can be submitted in any of the Patent Offices

Integrated Circuit Layout Designs



What is Layout Design?

- **Semiconductor Integrated Circuit** means a product having transistors and other circuitry elements, which are inseparably formed on a semiconductor material or an insulating material or inside the semiconductor material and designed to perform an electronic circuitry function.
- **Layout-Design** is the layout of transistors and other circuit elements, including lead wires connecting such elements and expressed in any manner in a semiconductor integrated circuit (IC).

Why protect Layout Designs?

- IC Layouts are creations of human mind;
- There is lot of investment of time and money in the creation but copying is very cheap;
- Fertile area with new circuit designs made every day to cater for miniaturization and novel applications.

Why Special protection?

- There may not be novelty so cannot be patented
- Copyright protection does not return the investments since commercial life of a design is limited.

Protection is against...

- Act of reproducing a layout design fully or in parts;
- Importing, selling or distributing commercially a protected layout design or IC incorporating it.

But identical design created *independently* by third party is *not prohibited*.

Layout-Design Registration

- Protection through registration
- Duration of registration: **Ten years**
- Criteria for Registration of a Chip Layout Design
 - Original,
 - Distinctive and
 - Capable of distinguishing from any other lay-out design.
 - *" Only the Layout-Design " - which essentially is the mask layout- floor planning of the integrated circuits can be registered under the SICLD Act 2000 and **not** the other information like any **idea, procedure, process, system, programme stored in the integrated circuit, method of operation etc.***

Layout-Design Registration

- Non-registrable Layout-designs
 - Not original
 - Has been commercially exploited anywhere in India or in a convention country
 - Not inherently distinctive
 - Not inherently capable of being distinguishable from any other registered layout-design.

Registration

- Office of the Semiconductor Integrated Circuits Layout-Design **Registry in Delhi**
- Under Ministry of Communications and Information Technology

Other IPRs

- **Trademarks** for goods and services used in trade and commerce
- **Geographical Indications** for goods having special qualities attributable to place of origin
- **Plant Varieties** protection for new plant varieties
- **Trade Secrets** protection for confidential information of commercial value (No separate law)

Using Biological Resources

- Needs to take note of the **Biological Diversity Act, 2002**
- “No person shall apply for any intellectual property right, ... **in or outside** India for any **invention** based on any research or information on a **biological resource obtained from India**, without obtaining the previous approval of the National Biodiversity Authority before making such applications.”[S.6(1)]
- Provided that if a person applies for a patent, permission of the NBA may be obtained after the acceptance of the patent but **before the sealing of the patent** by the patent authority concerned.
- Also applies to plant variety application

Using Biological Resources

- Approval of NBA is to be obtained before **Transfer of any Biological resources** of India or **results of research** to foreigners and non-resident Indians and foreign organisations
- Exception for **collaborative research** projects which conform to the policy guidelines issued by the MoEF and has the approval of the Central Government.

ADMINISTRATION OF IPRS

Administration of Rights

- IPRs are Private Property Rights
- Administered by owners
- Through Assignments and Licences
- Assignment is transfer of rights whereas licence is permission to do an act.

Administration of Rights

- Assignments
 - In existing and future work in case of copyright
 - Patents, Trade Marks, Designs and so on also
 - In writing
 - Identify work/invention/mark/design, specify rights, duration and territorial extent
 - Assignments of patents, trade marks, etc. to be registered with the office concerned
- Licences
- Transmission through testament of manuscript or artistic work includes copyright in that work.

Infringement

- Infringement is doing anything which is the exclusive right of the IP owner without his consent
- Permitting use of place for infringing communication to the public for profit
- Making, selling, distributing, displaying, importing of infringing copies, goods, etc.

Remedies for Infringement

- **Administrative**
 - Customs (Trademarks, copyrights, patents, geographical indications)
- **Civil (for all)**
 - Injunctions, Damages, Accounts
- **Criminal** (Copyright, Trademark, GI, Protection of Plant Varieties, and Layout-Design)
 - 6 months to 3 years imprisonment PLUS Rs. 50,000 to Rs. 2 lakh fine

Thank You for your attention